

CURRICULUM VITAE / FIELD-GUIDE EDITION

A FIELD GUIDE TO HOW ONE BUILDER WORKS

AI systems
&
humans

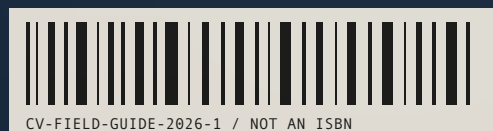
Field Guide of a Systems Builder

Michał Kałol — Data Science Manager & AI systems builder

Notes on experimentation integrity, content intelligence and
agent-readable knowledge systems.

Michał Kałol

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How to read this document

This is a CV in the shape of a **field guide** — ordered by what is most **defensible**, not most flattering. A machine sees a resume; a human sees field notes.

ROLE

Data Science
Manager & AI
systems builder

BASED

Warsaw, Poland /
Remote-first

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portfolio & field notes

I help people make progress when technical and organizational systems become too complex to hold in their heads.

The text is real, parseable text and the classic sections are intact — a plain ATS version is available on request. Written for founders, product leaders and AI teams looking for evidence of agency, systems thinking and honest self-assessment.

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Data scientist, product-minded manager, AI systems builder.

~15 years moving between research, web products, data science, small ventures and team leadership — mostly inside one large Polish media company at real product scale.

The common thread is not a tool or a title. It is a way of working: take a fuzzy, overloaded problem, find its structure, turn it into **work that can actually be done**, route most of it to people, and leave behind documentation and memory so the next person doesn't start from zero.

What separates me from a generic data scientist or a generic manager is **epistemic hygiene under pressure**. My most distinctive work is a documented, on-the-record self-retraction — a confident result I had written up at $p = 5 \times 10^{-6}$ that I traced to its sources and corrected to $p = 0.59$ (a null), after a stakeholder challenge prompted the audit — and the reusable discipline I built around it. **Most senior DS profiles can describe a win; far fewer can describe a confident result of their own that they killed before it could mislead anyone.**

WHAT I AM USUALLY GOOD AT

DIAGNOSING SYSTEMS

Seeing where **data, people, incentives and technology** fail to meet.

AMBIGUITY TO PROTOTYPE

Building the first working version **before the org finishes naming the problem.**

MAKING EXPERTS SCALABLE

Packaging knowledge so others use it **without months of onboarding.**

HONEST BOUNDARIES stated before an interviewer finds them: scale is solidly mid ($\sim 10^6$ -session datasets, $\sim 10^3$ -articles/day pipelines — not hyperscale); a meaningful part of my apparatus is single-operator (bus-factor-1); my dominant experimentation mode is reactive auditing, strong but not yet matched by a record of prospective, pre-registered design.

Capabilities, not buzzwords.

Online experimentation & measurement integrity

A/A validation; variance-inflation diagnosis (design-effect / intra-cluster correlation via clustered bootstrap); identity-instability confounds; false-positive auditing and self-retraction; delta-method SE correction; effective-sample-size checks; multiple-metric correction. My deepest, most authored vein.

A/A & A/B

DEFF / ICC

Clustered bootstrap

Error-surfacing

HONEST CEILING strongest reactively (catching what's wrong after the fact); I don't claim ownership of an experimentation platform, sequential / always-valid inference, or a portfolio of prospective designs.

Recommendation-system analytics & experimentation

I analyze, diagnose, tune and redesign reward functions for production contextual bandits (LinUCB / Thompson Sampling / OBPR-style) from raw event data, and run / audit their A/B tests. I build offline counterfactual evaluation: from-scratch LinTS / ridge proxies, MLP-Thompson via MC-Dropout, calibration and leakage controls.

Recsys analytics

Reward design

Offline / counterfactual eval

Live diagnosis

INTERVIEW BOUNDARY I am the analyst / diagnostician / offline implementer — not the production builder. Zero authored commits in the serving stack; my diagnoses end in a handoff to the owning team.

Content intelligence — credibility & quality evaluation

Machine evaluation of content quality and credibility — an interest with real research roots (web-content credibility, crowdsourced quality assessment) that now runs straight into LLM-as-arbiter content systems. One continuous 13-year thread: how do you machine-judge the quality and credibility of content?

Content quality

Credibility

LLM-as-arbiter

Crowdsourced eval

AI-augmented knowledge & workflow systems

LLM workflows, agents, **long-context agent-readable repositories**, RAG-like retrieval, trust boundaries, handoffs and — the part I am most deliberate about — **anti-confabulation discipline**: mandatory verification-before-write gates, adversarial solo LLM-role review, explicit fact / observation / interpretation / hypothesis tagging.

LLM workflows

Agents

Agent-readable context

Anti-confabulation

HONEST NOTE the value is operator discipline and integrative breadth, not a novel mechanism; a large part of this personal apparatus is single-operator.

Production LLM systems

End-to-end ownership of one live LLM pipeline: **streaming ingest → embed + score → LLM-arbiter → store**, with a nightly LLM curator, **deployed on Kubernetes via GitOps**, with the full design → deploy → debug → monitor loop.

Kafka → embed → arbiter

Kubernetes / GitOps

LLM curator

HONEST CEILING single-replica, no high-availability, throughput below its own target, no measured business impact yet (gated on an internal adoption demo), bus-factor ~ 1.

Technical leadership — player-coach & work-origination

Full formal people authority over a small specialist team (evaluate, hire, develop, set comp). I lead by originating and routing work: I **open ~2.1x more work than I'm assigned** and route **~78% of it outward** to a stable engineering cohort, keeping the spikes, PoCs and diagnoses.

Work-origination

Routing / delegation

Mentoring / onboarding

Player-coach

HONEST BOUNDARY the team is small (deliberately) — team-lead / player-coach altitude, not a Director-track large-org profile.

Large-scale media: data, experiments and AI.

Anonymized: a large Polish media company at real product scale — publishing, advertising, recommendation surfaces, online experimentation, high-volume behavioral data. Commercial details, internal architecture and confidential numbers are intentionally omitted.

Data Science Manager

MOST RECENT

Lead a small data science team across publishing and programmatic-advertising contexts, where analysis, ML, experimentation and product decisions meet.

- **Originate and route the team's work** across multiple projects; most of what I open is executed by a stable engineering cohort, while I take the spikes, PoCs and live diagnoses.
- Own **experimentation rigor**: A/A validation, variance-inflation / clustered-bootstrap diagnosis, identity-instability confounds, false-positive auditing.
- Diagnose, tune and **redesign reward functions** for production contextual bandits from raw event data; run / audit their A/B tests; route fixes to the owning teams.
- **Co-built and operate a production LLM** content-threading pipeline (streaming → embed / score → LLM-arbiter → store; GitOps deploy).
- Cross-project context-broker; report into a senior-director-adjacent line; co-set and negotiate bonus-linked goals with executive stakeholders.

Data Science Team Leader

PRIOR

Led and coordinated data science across multiple product and business contexts; moved the team's practice from isolated analysis toward **production-ready workflows**; drove the team's R-to-Python migration; onboarded DS hires (stack → usage → architecture → route to specialist), including enabling a non-traditional hire to contribute to production.

Data Scientist

EARLIEST

NLP, topic modeling, classification and linked-data / knowledge-base work; built practical data-gathering and processing pipelines for messy web-scale inputs.

Field note on terrain. Large traffic and a real product, lean teams against high expectations, fast decisions, dependence on other overloaded teams — this shaped how I work. It rewards a person who can be analyst, manager, diagnostician, translator and firefighter at once.

Experimentation as an error-surfacing system.

Good experimentation is not about never being wrong. It is about **making wrongness visible early enough** that nobody ships it.

The signature story. On an internal A/B / A/A analysis I produced a confident, correction-passing result at $p = 5 \times 10^{-6}$ and had already written it into a one-pager. Auditing at the raw-count level, I found three stacked mistakes — a chart-title-vs-dataset mismatch, a cumulative-sum inflation that $\sim 6\times$ d the denominator, and a continuous metric needing a Welch-type correction — and corrected the true value to $p = 0.59$: a null. I then turned the incident into reusable policy: a **verification checklist** and a **banned-overclaim lexicon**.

HONESTY I KEEP IN THE STORY the audit was prompted by a stakeholder challenge, not self-caught before publication. Excellent recovery and codified prevention — presented as exactly that, not flawless foresight.

Supporting evidence (all authored, on real data)

- **Clustered variance — my standout.** Empirical **design-effect $\sim 12.5 / 18.5$** and **ICC ~ 0.40** , via clustered bootstrap-by-user on **$\sim 1.5\text{M}$ sessions / $\sim 163\text{k}$ users** — correcting my own back-of-envelope estimate.
- **The identity-instability paradox.** Removing "flip-flop" users **increased** A/A variance ($\sim 1.77 \rightarrow 3.71$) — heavy users stabilize buckets by the law of large numbers: "flips add noise, not bias."
- **A delta-method SE catch** that had inflated a variance-rate standard error by $\sim 76\times$.
- **A Simpson's-paradox confound** where an aggregate effect reversed under subgroup decomposition; I confirmed a stakeholder's hypothesis while correcting its mechanism ("right direction, wrong cause").
- **Effective-sample-size diagnosis** ($12\text{--}250\times$ below nominal on heavy-traffic days; bot / scraper contamination passing naive filters).

HONEST CEILING no sequential / always-valid inference in my record (fixed-horizon by design); no executed off-policy-evaluation number I ran end-to-end; causal-clustering correction left as a named open item. I state these before being asked.

AIP Workspace: **organizational memory** as agent context.

A private internal field experiment testing one thesis: an organization's scattered, stale, expert-locked knowledge can be compressed into a form an **AI agent can actually use** — to lower cognitive load and let people work more independently inside a complex stack.

PROBLEM OBSERVED

Complex stacks punish newcomers and create **hidden dependence on a few experts**; documentation exists but is scattered, stale, or absent at the moment of work.

HYPOTHESIS

If context, decisions, code knowledge and operational notes live in an **agent-readable workspace**, the system reduces cognitive load and accelerates independent work.

MECHANISM

Analyses, handoffs, operational recipes, mirrors of org sources and reusable agent skills — with explicit trust boundaries, human review and a **mandatory verification gate** before anything becomes shared canon.

EARLY SIGNAL (QUALITATIVE)

People who had struggled with parts of the stack began **moving further on their own**; the gain was safer knowledge transfer and lower cognitive load, not raw code generation.

HONEST CAVEATS, UP FRONT partly aspirational and largely single-operator (bus-factor ~ 1); no quantitative adoption metric; in at least one logged session its retrieval layer went unqueried — an artifact's presence is not the same as its changing behavior. I name this because the discipline of the workspace is the point.

The interesting unit is not the prompt. It is the workspace: the durable, agent-readable context that turns an LLM from a clever assistant into an organizational tool.

Public proof of building.

Some professional, some personal, some unfinished — together they show one pattern repeated across unrelated domains: **build a system, not a one-off**; leave artifacts and memory behind.

01 Rana / The Wound

A finished, **print-ready hard-SF book** (~48k words) written AI-native: multi-part structure, iterative adversarial critique rounds, multi-language builds (ePub / PDF / HTML), consistency-checking tooling. **Signal:** long-form thinking, editing discipline, human-AI collaboration, the ability to finish.

02 Furki

A networked arcade game (authoritative server, behavior-tree + LLM-driven bots, AI-generated assets); a **playable PvP MVP**. **Signal:** product play, game mechanics, codebase exploration, sustained side work.

03 GizmoZero

A robotics prototype (single-board computer, motor drivers, sensors, camera, LLM integration) built with my kids; honestly unfinished. **Signal:** physical-computing curiosity, crossing the software/hardware boundary despite messy constraints.

04 AIP Workspace

The internal AI-augmented knowledge system above; private, so described as a case study rather than shown. **Signal:** **applied agentic AI**, organizational memory, workflow design, adoption discipline.

Before the AI boom: web, ventures, research.

EPI2

Co-founded a small **data-science consultancy** delivering NLP solutions to external clients.

OSEMECZKA

Founded a service around data entry, forms, OCR / OMR workflows, validation and operational tooling.

EARLIER WEB

Web development and design, including localization / marketing-communications for enterprise clients.



Research roots

I started in **web-content credibility** and crowdsourced quality assessment at the Polish-Japanese Academy of Information Technology (social informatics / web mining) — building corpora, models and publications — **before it became fashionable in the era of AI-generated content**. A small but real body of work (~9 papers; one flagship credibility-corpus paper well-cited and still accruing citations), active mainly ~2013–2018.

STATED HONESTLY these are research roots, not academic authority: not an active researcher (no publications since ~2018) and I don't lead with a citation count. The value is the continuity — credibility / quality evaluation → recommendation quality & A/B testing → LLM-arbiter content systems — one thread, ~13 years.

Education

SOCIAL INFORMATICS

PhD Candidate · PJAiT

COMPUTER SCIENCE

MSc · PJAiT

QUANTITATIVE METHODS

BSc · Warsaw School of Economics

Tools are listed last on purpose.

The strongest signal is not a framework — it is combining tools into a working system.

Programming

Python, SQL (ClickHouse-heavy), JavaScript / TypeScript, HTML / CSS, Bash; reading-fluent Go; some R

Data & ML

pandas / Polars, NumPy, PySpark; recommendation-system analytics, contextual-bandit diagnostics & offline evaluation, embeddings, NLP, calibration

Experimentation / stats

A/B & A/A, clustered bootstrap, DEFF / ICC, effective-sample-size, delta-method SE, multiple-metric correction, CUPED-awareness

Data stores & infra

Postgres, ClickHouse, Kafka, Redis, Neo4j, Qdrant, Elasticsearch, Druid; Kubernetes + ArgoCD (GitOps), Docker, Airflow, Spark

AI systems

LLMs (multi-model by role / cost), AI agents, MCP / agent tooling, RAG-like retrieval, knowledge graphs, local-model serving (POC-grade), human-in-the-loop design

Leadership

small-team people management (hire / develop / comp), work-origination & routing, mentoring, stakeholder alignment, technical quality-gating

Languages

POLISH

Native

ENGLISH

Full professional

GERMAN / SPANISH

Intermediate / rudimentary

For the right reader.

For teams that value people who can build across boundaries: data and product, research and business, AI and workflow, prototype and organization.

GOOD FIT

AI-native / applied-LLM product companies; lean **senior / staff "lead DS/ML"** roles without a heavy platform team, where vertical ownership and work-origination are the job; experimentation-integrity, content-intelligence and internal-AI / knowledge-systems functions; a small builder-team lead where hands-on credibility and routing — not headcount — are valued.

POOR FIT

Slide-only "AI transformation," reporting-first cultures, consulting factories; a pure-management Director role at a large, politically heavy org; a mature platform org where individual rigor becomes invisible table stakes; any role needing a **production contextual-bandit / recsys builder** with a shipped-lift track record — that is honestly not me, and I won't claim it.

I am most useful where there is real complexity, real users, real data and enough trust to build the system properly — and where being right matters more than sounding right.

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